From Work Integrals to Phase-Space Area and Action Variables

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An integral measures accumulation, but the geometry of what is accumulated matters. Work is integrated along an oriented path, and the same line-integral mechanism reappears when one measures the area enclosed by a periodic trajectory in phase space. In that setting the integral $\oint p\,dq$ turns out to determine the oscillation period without solving the equation of motion explicitly. This is a concrete route from first-year calculus to Hamiltonian mechanics.[1]

§1.1 Work is the accumulation of force along a path

In one dimension, when a force $F(x)$ acts while a particle moves from a to b,

$$W = \int_a^b F(x)\,dx.$$

For example, the spring force $F(x) = -kx$ performs work

$$W_{a\to b} = -\frac{k}{2}(b^2 - a^2).$$

This calculation naturally suggests the potential energy $U(x) = kx^2/2$, for which $F = -U'$ and $W = U(a) - U(b)$.

For a curve $\gamma(t) = (x(t), y(t), z(t))$ in space, the displacement is $d\mathbf{r} = \gamma'(t)\,dt$. The signed work becomes

$$\boxed{W_\gamma = \int_\gamma F \cdot d\mathbf{r} = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\,dt.}$$

It depends on the path's orientation. If $F = -\nabla U$, the chain rule gives

$$F(\gamma(t)) \cdot \gamma'(t) = -\frac{d}{dt} U(\gamma(t)),$$

so

$$\boxed{W_\gamma = U(\gamma(a)) - U(\gamma(b)).}$$

For a conservative force, the work depends only on the endpoints and vanishes around every closed loop.

§1.2 Circulation converts a line integral to area

Consider the planar field $F = (-y, x)$, which rotates counterclockwise around the origin. On a circle of radius R,

$$\gamma(t) = (R\cos t, R\sin t), \quad 0 \le t \le 2\pi,$$

[1] MIT 18.02 (line integrals and Green's theorem); MIT 8.09, Classical Mechanics III (action variables); V. I. Arnold, *Mathematical Methods of Classical Mechanics*.

one obtains

$$\int_\gamma F \cdot d\mathbf{r} = \int_0^{2\pi} R^2 \, dt = 2\pi R^2.$$

The curl of this field is constant:

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 2.$$

Green's theorem therefore explains the result as

$$\oint_{\partial D} (-y \, dx + x \, dy) = \iint_D 2 \, dx \, dy = 2 \text{ area}(D).$$

The calculation is exact for any sufficiently regular positively oriented planar region D, not just circles.

Choosing instead the one-form $x \, dy$, whose exterior derivative is $dx \wedge dy$, gives the particularly simple area formula

$$\boxed{\text{area}(D) = \oint_{\partial D} x \, dy = -\oint_{\partial D} y \, dx.}$$

A planar region can therefore be measured by integrating around its boundary. This is the bridge from accumulation along a path to geometric area.

§1.3 The canonical one-form on phase space

Now consider a particle moving in one dimension, with position q, momentum $p = m\dot{q}$, and mechanical energy

$$H(q, p) = \frac{p^2}{2m} + V(q).$$

The pair (q, p) describes a point in *phase space*. A periodic mechanical motion traces a closed curve there. Along this curve, define the integral

$$\boxed{J(E) = \oint_{H=E} p \, dq.}$$

Unlike the mechanical work $\int F \, dq$, this integral measures momentum against displacement. Its units are momentum times length, or action.

The one-form $\vartheta = p \, dq$ is the canonical one-form in these coordinates. Its differential is

$$d\vartheta = dp \wedge dq = -dq \wedge dp.$$

Green's theorem thus interprets $J(E)$ as an oriented phase-space area. For the usual Hamiltonian evolution of a one-dimensional oscillation, the energy orbit is traversed clockwise in (q, p)-coordinates; the integral $J(E)$ equals the ordinary positive area enclosed by that orbit.

§1.4 Action of the Harmonic Oscillator

Take $V(q) = kq^2/2$ with $\omega = \sqrt{k/m}$. The energy curve is

$$\frac{p^2}{2mE} + \frac{kq^2}{2E} = 1.$$

It is an ellipse with horizontal semiaxis $q_{\max} = \sqrt{2E/k}$ and vertical semiaxis $p_{\max} = \sqrt{2mE}$. Its area is

$$J(E) = \pi q_{\max} p_{\max} = \pi \sqrt{\frac{2E}{k}} \sqrt{2mE} = \boxed{\frac{2\pi E}{\omega}}.$$

The same answer can be checked by following one orbit:

$$q(t) = A\cos\omega t, \qquad p(t) = -m\omega A \sin\omega t, \qquad E = \frac{1}{2} m\omega^2 A^2.$$

Since $dq/dt = -A\omega \sin\omega t$,

$$\oint p\, dq = m\omega^2 A^2 \int_0^{2\pi/\omega} \sin^2 \omega t\, dt = \pi m \omega A^2 = \frac{2\pi E}{\omega}.$$

The boundary-integral interpretation and the direct time integral are equivalent because the particle traces precisely the ellipse.

§1.5 Differentiating enclosed area recovers the period

The oscillator identity is an instance of a general fact. Suppose

$$H(q,p) = \frac{p^2}{2m} + V(q)$$

has a regular family of periodic energy orbits between simple turning points $q_-(E) < q_+(E)$. On the upper half of the orbit,

$$p_+(q,E) = \sqrt{2m(E - V(q))},$$

and the lower half has momentum $-p_+$. The enclosed area is therefore

$$J(E) = 2\int_{q_-(E)}^{q_+(E)} \sqrt{2m(E - V(q))}\, dq.$$

Differentiating with respect to energy is legitimate away from critical energy levels. The integrand vanishes at the moving turning points, so their endpoint contributions vanish. We obtain

$$\begin{aligned} \frac{dJ}{dE} &= 2\int_{q_-}^{q_+} \frac{m}{\sqrt{2m(E - V(q))}}\, dq \\ &= 2\int_{q_-}^{q_+} \frac{dq}{\dot q}. \end{aligned}$$

Here $\dot q = p/m > 0$ on the upper branch. The last integral is twice the travel time from the left turning point to the right turning point, hence the full period:

$$\boxed{\frac{dJ}{dE} = T(E).}$$

For the harmonic oscillator $J(E) = 2\pi E/\omega$, this immediately yields $T = 2\pi/\omega$, without solving $m\ddot q + kq = 0$.

The formula is useful for anharmonic potentials: the period's energy dependence is encoded in the changing area of the energy contours. If an energy approaches a separatrix, the period may diverge because motion slows near an unstable equilibrium; the derivative of the phase-space area captures that divergence.

§1.6 Canonical Coordinate Changes

The phase-space area form is $\omega = dq \wedge dp$. A canonical change of coordinates $(q, p) \mapsto (Q, P)$ preserves this two-form:

$$dq \wedge dp = dQ \wedge dP.$$

Locally, the difference between the canonical one-forms is therefore closed:

$$d(p\,dq - P\,dQ) = 0.$$

On a simply connected coordinate patch it is exact, say $p\,dq - P\,dQ = dS$. Around a closed orbit lying in such a patch,

$$\oint p\,dq - \oint P\,dQ = \oint dS = 0.$$

Consequently the action is a coordinate-invariant measure of the enclosed symplectic area. The requirement about the coordinate patch matters: a closed form on a multiply connected domain need not have a globally defined potential.

The path integral has now done three kinds of work. Integrating $F\,dq$ gives the mechanical work performed by a force, integrating around the boundary of a planar region measures ordinary area, and integrating $p\,dq$ around a periodic trajectory measures phase-space area. In a one-degree-of-freedom Hamiltonian system, differentiating this last area with respect to energy gives a directly observable quantity: the period.